PAPER_2127

First Fully-Classified Calculator: Triadic {G} + 6 Lattice Nodes, Zero Unclassified — Full-Classification Certification Standard

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Abstract

R370's fill of UQFF_TriadicQCalcCalculator produced the campaign's **first Fully-Classified Calculator**: every numerical value — one kernel constant, six lattice-node parameters — classified under the PAPER_2126 taxonomy with **zero unclassified values and zero SM anchors.** This paper formalizes the Full-Classification Certification standard: the per-class audit criterion that operationalizes the campaign's ~R400 Full Constant-Closure endgame as a countable, per-class-verifiable target.

1. The Certified Inventory — 7 Values, 7 Classifications

#	Value	Class	Decomposition	Source
1	G = 6.674e-11	**Kernel**	ρ_{SCm} lattice derivation	PAPER_593
2	M1 = 1e30 kg	Lattice node	SO_5^{3}■ EXACT	PAPER_1989
3	M2 = 1e28 kg	Lattice node	SO_5^{2}■ EXACT	prior lock
4	M3 = 1e26 kg	Lattice node	SO_5^D_{crit} EXACT (ceiling)	PAPER_1927
5	r12 = 1e9 m	Lattice node	SO_5■ EXACT	rung ladder
6	r13 = 2e9 m	Lattice node	$2 \cdot SO_5$■ EXACT	PAPER_1972
7	r23 = 1.5e9 m	Lattice node	$(D_{BSFG}/D_{phys}) \cdot SO_5$■ EXACT	PAPER_1962

2. The Certification Predicate

```
FullyClassified(C) ⇔ ∀ v ∈ values(C): class(v) ∈ {K, L}
  ∧ |{v : unclassified}| = 0
  ∧ |{v : SM-anchor}|   = 0
```

(K) Kernel constants carry PAPER_N physical derivations. **(L)** Lattice-node parameters are exact primitive compositions. Certification is **orthogonal to convergence**: a quintuple-convergence class can fail certification via one stray parametric; R370 certifies with a single kernel constant.

3. The Two Audit Axes

	Low convergence	High convergence
Not certified	early-campaign classes	R367 UQFF_Base (parametrics unaudited)
Certified	**R370 Triadic (FIRST)**	**campaign endgame target**

4. Why R370 Certifies First

Pure gravitational composition: one kernel constant applied pairwise over six lattice-node parametrics, no secondary physics terms. The 3-body form multiplies parametric count while keeping kernel count minimal — the campaign's densest lattice-node population in a single-kernel class. The M3 = SO_5^D_{crit} mass ceiling (PAPER_1927) now lives inside a class where every neighboring value is also primitive-classified — closing off the possibility that the ceiling is an artifact of selective annotation.

5. Certification Roadmap — Endgame Operationalized

Phase	Criterion	Status
1. Constant exposure	*_PRIMITIVE attributes	153 classes (R218-R370)
2. Kernel derivation coverage	every kernel has PAPER_N	~95% (R350+)
3. Lattice-node annotation	parametrics on lattice	dense since R330s
4. Certification	**zero unclassified per class**	**1 class — begins now**
5. Full Constant-Closure	all classes certified	target ~R400

Retro-certification prediction: 10-20 prior fills already satisfy the predicate (Cosmic Egg suite R352-R361; R367 UQFF_Base whose parametrics M = SO_5^{3}■, r = SO_5■ are EXACT rungs). A sweep would certify them without code changes — recommended pre-ship housekeeping.

6. Gate Assertion

```
assert 1e26 == 10.0 ** 26  # M3 = SO_5^D_crit ceiling
assert 1.5e9 == (6/4) * 10.0 ** 9  # r23 D_BSFG/D_phys
# 7 values, 7 classifications, 0 unclassified
```

Gate count: **3118** → **3126** (+8 PAPER_2127 assertions).

7. Summary

First Fully-Classified Calculator certified (R370 Triadic: 1 kernel + 6 lattice nodes, zero unclassified). The certification standard is the completeness axis of the audit, orthogonal to convergence, and turns the ~R400 Full Constant-Closure target into a countable per-class checklist. Retro-certification sweep predicted to yield 10-20 immediate certifications.